2024-06-19 Crystal field equations with defects using tesseral harmonics

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I. TESSERAL HARMONICS

Spherical harmonics are complex functions. When used in the crystal field calculations we of course always have a purely real energy in the end. So it is convenient to express everything in tesseral harmonics, which are purely real combinations of spherical harmonics. A good source for the relationship is https://quanty.org/physics_chemistry/orbitals/z

$$Z_{l,m} = \begin{cases} Y_{lm} & m = 0 \\ \frac{1}{\sqrt{2}} \left(Y_{l,-m} + (-1)^m Y_{l,m} \right) & m > 0 \\ \frac{\mathrm{i}}{\sqrt{2}} \left(Y_{l,m} - (-1)^m Y_{l,-m} \right) & m < 0 \end{cases} \tag{1}$$

where it should be understood that the m factors are consistent with the $m > 0$ and $m < 0$. i.e. in the case when $m < 0$ such as $m = -1$ we would have

$$Z_l^{(1)} = \frac{i}{\sqrt{2}} \left(Y_l^{(-1)} - (-1)^{(-1)} Y_l^{(-(-1))} \right) = \frac{i}{\sqrt{2}} \left(Y_l^{(-1)} - (-1)^{(-1)} Y_l^{(1)} \right) \tag{2}$$

The first 6 orders of tesseral harmonics are written on p164 of Birss - Symmetry and Magnetism. There is also the identities

$$Y_l^{(m)} = (-1)^{-m} \left(Y_l^{(-m)} \right)^* \tag{3}$$

$$Y_l^{(-m)} = (-1)^m \left(Y_l^{(m)} \right)^* \tag{4}$$

$$\left(Y_l^{(m)} \right)^* = (-1)^{-m} Y_l^{(-m)} \tag{5}$$

$$\left(Y_l^{(-m)} \right)^* = (-1)^m Y_l^{(m)} \tag{6}$$

So we can also write the tesseral harmonics as

$$Z_l^{(m)} = \begin{cases} Y_l^0 & m = 0 \\ \frac{1}{\sqrt{2}} \left[(-1)^m \left(Y_l^{(m)} \right)^* + \left(Y_l^{(-m)} \right)^* \right] & m > 0 \\ \frac{\mathrm{i}}{\sqrt{2}} \left[(-1)^{-m} \left(Y_l^{(-m)} \right)^* - (-1)^m (-1)^m \left(Y_l^{(m)} \right)^* \right] & m < 0. \end{cases} \tag{7}$$

This simplifies because both $(-1)^m(-1)^m = (-1)^{2m} = 1$ and $(-1)^{-m} = (-1)^m$ if m is integer. Giving

$$Z_l^{(m)} = \begin{cases} Y_l^0 & m = 0 \\ \frac{1}{\sqrt{2}} \left[(-1)^m \left(Y_l^{(m)} \right)^* + \left(Y_l^{(-m)} \right)^* \right] & m > 0 \\ \frac{\mathrm{i}}{\sqrt{2}} \left[(-1)^m \left(Y_l^{(-m)} \right)^* - \left(Y_l^{(m)} \right)^* \right] & m < 0. \end{cases} \tag{8}$$

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II. CRYSTAL FIELD PERTURBATION

Chris defines the crystal field perturbation as (Eq. 2 in their main text)

$$\Delta B_{lm} = -\langle r^l \rangle \left(\frac{4\pi}{2l+1} \right)^{1/2} \frac{\Delta Z_J}{R_J^{l+1}} Y^*_{lm}(\hat{\mathbf{R}}_J) \tag{9}$$

and in Fig. 1 they comment that $\Delta B_{lm} = \Delta B_{l-m}$. The m indices always come in $+/-$ pairs. We can rewrite the tesseral Harmonics in terms of the conjugated spherical harmonics

$$\left(Y_l^{(m)} \right)^* = \begin{cases} Z_l^{(0)} & m = 0 \\ (-1)^m \left[\sqrt{2} Z_l^{(m)} - \left(Y_l^{(-m)} \right)^* \right] & m > 0 \\ (-1)^m \left[\mathrm{i}\sqrt{2} Z_l^{(m)} - \left(Y_l^{(-m)} \right)^* \right] & m < 0. \end{cases} \tag{10}$$

III. CRYSTAL FIELD

From Chris' paper the expression for the crystal field can be written in general as

$$\mathscr{H}_{\mathrm{CF}} = \sum_{lm} \mathcal{A}_l d_l^{0m}(\Theta) e^{im\phi} B_{lm} \tag{11}$$

We first clarify the indices on the summation

$$\mathscr{H}_{\mathrm{CF}} = \sum_{l=2,4,6} \sum_{m=-l}^{l} \mathcal{A}_l d_l^{0m}(\Theta) e^{im\phi} B_{lm} \tag{12}$$

where we usually only have the first 3 terms in l (and odd terms are forbidden by symmetry). Chris defines

$$d_l^{0m}(\Theta) = \sqrt{\frac{(l-m)!}{(l+m)!}} P_l^m(\cos\Theta) = (-1)^m d_l^{0-m}(\Theta) \tag{13}$$

So we rewrite as

$$\mathscr{H}_{\mathrm{CF}} = \sum_{l=2,4,6} \sum_{m=-l}^{l} \sqrt{\frac{(l-m)!}{(l+m)!}} P_l^m(\cos\Theta) e^{im\phi} B_{lm} \tag{14}$$

We can recognise the Schmidt semi-normalized (or Racah's normalized) spherical harmonics (see https://en.wikipedia.org/wiki/Spherical_harmonics):

$$Y_{l,m}(\Theta,\phi) = \sqrt{\frac{(l-m)!}{(l+m)!}} P_l^m(\cos\Theta) e^{im\phi} \tag{15}$$

hence

$$\mathscr{H}_{\mathrm{CF}} = \sum_{l=2,4,6} \sum_{m=-l}^{l} Y_{l,m}(\Theta,\phi) B_{l,m} \tag{16}$$

We can then rewrite this in tesseral harmonics so that everything is purely real. So we will have an expansion

$$\mathscr{H}_{\mathrm{CF}} = \sum_{l=2,4,6} \sum_{m=-l}^{l} Z_{l,m}(\Theta,\phi) C_{l,m} \tag{17}$$

where

$$C_{l,m} = \begin{cases} B_{l0} & m = 0 \\ \frac{1}{\sqrt{2}} \left(Y_{l,-m} + (-1)^m Y_{l,m} \right) & m > 0 \\ \frac{\mathrm{i}}{\sqrt{2}} \left(Y_{l,m} - (-1)^m Y_{l,-m} \right) & m < 0 \end{cases} \tag{18}$$

Of course, for efficiency, we want to use the Cartesian spin components rather than angles. So we would use

$$\mathscr{H}_{\mathrm{CF}} = \sum_{l=2,4,6} \sum_{m=-l}^{l} Y_l^m(\mathbf{S}) B_{lm} \tag{19}$$

There should be some relationship between the crystal field coefficients B_{lm} which allows us to rewrite this all in tesseral harmonics and have a purely real function. But let's keep things in complex form for now and optimise this later. We next need the local effective field,

$$\mathbf{H}_i = -\frac{1}{\mu_i}\frac{\partial \mathscr{H}}{\partial \mathbf{S}_i} \tag{20}$$

so

$$\mathbf{H}_i = -\frac{1}{\mu_i} \sum_{l=2,4,6} \sum_{m=-l}^{l} \frac{\partial}{\partial \mathbf{S}_i}\Big(Y_l^m(\mathbf{S}_i)\Big) B_{lm} \tag{21}$$

The differentials are a little complicated to generate in general (see Bigi *et al.*, 2023) due to the potential for divergences at the poles. However, if we use the identity $Y_l^m = (-1)^{-m}\left(Y_l^{-m}\right)^*$ then we actually only need 15 equations to be able to implement all of the terms in the sum (because l is limited to $2, 4, 6$). Hence it's probably simplest and most efficient to hard code the expressions we need.

We also need to be careful to perform the differentiation correctly because $\mathbf{S}$ must be a unit vector. Hence we either need to explicitly include the normalisation factor, or differentiate the spherical form and the convert to Cartesian. The latter is probably simplest because Mathematica implements the spherical harmonics in spherical form.

The results from Mathematica are:

$$-\nabla_{\mathbf{s}} Y_2^0(\mathbf{s}) = \begin{pmatrix} 3S_x S_z^2 \\ 3S_y S_z^2 \\ 3S_z\left(S_z^2 - 1\right) \end{pmatrix} \tag{22}$$

$$-\nabla_{\mathbf{s}} Y_2^1(\mathbf{s}) = \begin{pmatrix} \sqrt{\frac{3}{2}} S_z\left(S_z^2 - (S_x + iS_y)^2\right) \\ i\sqrt{\frac{3}{2}} S_z\left(S_z^2 + (S_x + iS_y)^2\right) \\ \sqrt{\frac{3}{2}}\left(1 - 2S_z^2\right)(S_x + iS_y) \end{pmatrix} \tag{23}$$

$$-\nabla_{\mathbf{s}} Y_2^2(\mathbf{s}) = \begin{pmatrix} i\sqrt{\frac{3}{2}}(S_x + iS_y)\left(S_x S_y + i\left(S_y^2 + S_z^2\right)\right) \\ \sqrt{\frac{3}{2}}(S_y - iS_x)\left(iS_x S_y + S_x^2 + S_z^2\right) \\ \sqrt{\frac{3}{2}} S_z (S_x + iS_y)^2 \end{pmatrix} \tag{24}$$

$$-\nabla_{\mathbf{s}} Y_4^0(\mathbf{s}) = \begin{pmatrix} \frac{5}{2} S_x S_z^2 \left(7S_z^2 - 3\right) \\ \frac{5}{2} S_y S_z^2 \left(7S_z^2 - 3\right) \\ \frac{5}{2} S_z \left(S_z^2 - 1\right) \left(7S_z^2 - 3\right) \end{pmatrix} \tag{25}$$

$$-\nabla_{\mathbf{s}} Y_4^1(\mathbf{s}) = \begin{pmatrix} \frac{1}{4}\sqrt{5} S_z \left(-S_z^2 \left(S_x + iS_y\right)\left(21 S_x + iS_y\right) + 3\left(S_x - iS_y\right)\left(S_x + iS_y\right)^3 + 4S_z^4\right) \\ \frac{1}{4}\sqrt{5} S_z \left(iS_z^2 \left(S_x + iS_y\right)\left(S_x + 21 i S_y\right) + 3\left(-S_y + iS_x\right)^3\left(S_x - iS_y\right) + 4iS_z^4\right) \\ -\frac{1}{4}\sqrt{5}\left(28 S_z^4 - 27 S_z^2 + 3\right)\left(S_x + iS_y\right) \end{pmatrix} \tag{26}$$

$$-\nabla_{\mathbf{s}} Y_4^2(\mathbf{s}) = \begin{pmatrix} \frac{1}{2}\sqrt{\frac{5}{2}}\left(S_y - iS_x\right)\left(iS_z^2\left(S_x + iS_y\right)\left(8S_x + 5iS_y\right) + S_y\left(S_x - iS_y\right)\left(S_x + iS_y\right)^2 - 6iS_z^4\right) \\ \frac{1}{2}\sqrt{\frac{5}{2}}\left(S_x + iS_y\right)\left(S_z^2\left(5S_x + 8iS_y\right)\left(S_y - iS_x\right) + S_x\left(S_x + iS_y\right)^2\left(S_y + iS_x\right) - 6iS_z^4\right) \\ \sqrt{\frac{5}{2}} S_z \left(7S_z^2 - 4\right)\left(S_x + iS_y\right)^2 \end{pmatrix} \tag{27}$$

$$-\nabla_{\mathbf{s}} Y_4^3(\mathbf{s}) = \begin{pmatrix} -\frac{1}{4}\sqrt{35} S_z \left(S_x + iS_y\right)^2\left(4iS_xS_y + S_x^2 - 3\left(S_y^2 + S_z^2\right)\right) \\ \frac{1}{4} i\sqrt{35} S_z \left(S_x + iS_y\right)^2 \left(4iS_xS_y + 3S_x^2 - S_y^2 + 3S_z^2\right) \\ \frac{1}{4}\sqrt{35}\left(1 - 4S_z^2\right)\left(S_x + iS_y\right)^3 \end{pmatrix} \tag{28}$$

$$-\nabla_{\mathbf{s}} Y_4^4(\mathbf{s}) = \begin{pmatrix} \frac{1}{2} i\sqrt{\frac{35}{2}}\left(S_x + iS_y\right)^3\left(S_xS_y + i\left(S_y^2 + S_z^2\right)\right) \\ \frac{1}{2}\sqrt{\frac{35}{2}}\left(-S_y + iS_x\right)^3\left(iS_xS_y + S_x^2 + S_z^2\right) \\ \frac{1}{2}\sqrt{\frac{35}{2}} S_z \left(S_x + iS_y\right)^4 \end{pmatrix} \tag{29}$$

$$-\nabla_{\mathbf{s}} Y_6^0(\mathbf{s}) = \begin{pmatrix} \frac{21}{8} S_x S_z^2 \left(33 S_z^4 - 30 S_z^2 + 5\right) \\ \frac{21}{8} S_y S_z^2 \left(33 S_z^4 - 30 S_z^2 + 5\right) \\ \frac{21}{8} S_z \left(S_z^2 - 1\right) \left(33 S_z^4 - 30 S_z^2 + 5\right) \end{pmatrix} \tag{30}$$

$$-\nabla_{\mathbf{s}} Y_6^1(\mathbf{s}) = \begin{pmatrix} \frac{1}{8}\sqrt{\frac{21}{2}} S_z \left(-4 S_z^4 \left(S_x + i S_y\right)\left(25 S_x - 3 i S_y\right) + 5 S_z^2 \left(S_x - i S_y\right)\left(S_x + i S_y\right)^2 \left(17 S_x + 3 i S_y\right) - 5 \left(S_x - i S_y\right)^2 \left(S_x + i S_y\right)^4 \right. \\ \frac{1}{8}\sqrt{\frac{21}{2}} S_z \left(4 S_z^4 \left(3 S_x - 25 i S_y\right)\left(S_y - i S_x\right) + 5 S_z^2 \left(S_x - i S_y\right)\left(S_x + i S_y\right)^2 \left(17 S_y - 3 i S_x\right) + 5 i \left(S_x - i S_y\right)^2 \left(S_x + i S_y\right)^4 - \right. \\ -\frac{1}{8}\sqrt{\frac{21}{2}} \left(198 S_z^6 - 285 S_z^4 + 100 S_z^2 - 5\right)\left(S_x + i S_y\right) \end{pmatrix} \tag{31}$$

$$-\nabla_{\mathbf{s}} Y_6^2(\mathbf{s}) = \begin{pmatrix} \frac{1}{16}\sqrt{105}\left(S_x + i S_y\right)\left(64 S_x S_z^4 \left(S_x + i S_y\right) - S_z^2 \left(S_x - i S_y\right)\left(S_x + i S_y\right)^2 \left(19 S_x + 15 i S_y\right) + i S_y \left(S_x - i S_y\right)^2 \left(S_x + i S_y\right)^3 \right. \\ \frac{1}{16}\sqrt{105}\left(S_y - i S_x\right)\left(64 i S_y S_z^4 \left(S_x + i S_y\right) - S_z^2 \left(S_x + i S_y\right)^2 \left(4 i S_x S_y + 15 S_x^2 + 19 S_y^2\right) + S_x \left(S_x - i S_y\right)^2 \left(S_x + i S_y\right)^3 - \right. \\ \frac{1}{16}\sqrt{105} S_z \left(99 S_z^4 - 102 S_z^2 + 19\right)\left(S_x + i S_y\right)^2 \end{pmatrix} \tag{32}$$

$$-\nabla_{\mathbf{s}} Y_6^3(\mathbf{s}) = \begin{pmatrix} \frac{3}{16}\sqrt{105} S_z \left(S_x + i S_y\right)^2 \left(-S_z^2 \left(S_x + i S_y\right)\left(13 S_x + 5 i S_y\right) + \left(S_x - i S_y\right)\left(S_x + i S_y\right)^2 \left(S_x + 3 i S_y\right) + 8 S_z^4\right) \\ -\frac{3}{16} i \sqrt{105} S_z \left(S_x + i S_y\right)^2 \left(-S_z^2 \left(S_x + i S_y\right)\left(5 S_x + 13 i S_y\right) + \left(S_x + i S_y\right)^2 \left(-2 i S_x S_y + 3 S_x^2 + S_y^2\right) - 8 S_z^4\right) \\ -\frac{3}{16}\sqrt{105}\left(22 S_z^4 - 15 S_z^2 + 1\right)\left(S_x + i S_y\right)^3 \end{pmatrix} \tag{33}$$

$$-\nabla_{\mathbf{s}} Y_6^4(\mathbf{s}) = \begin{pmatrix} \frac{3}{8}\sqrt{\frac{7}{2}}\left(S_x + i S_y\right)^3 \left(S_z^2 \left(S_x + i S_y\right)\left(13 S_x + 18 i S_y\right) - 2 i S_y \left(S_x - i S_y\right)\left(S_x + i S_y\right)^2 - 20 S_z^4\right) \\ \frac{3}{8} i \sqrt{\frac{7}{2}}\left(S_x + i S_y\right)^3 \left(-S_z^2 \left(S_x + i S_y\right)\left(18 S_x + 13 i S_y\right) + 2 S_x \left(S_x - i S_y\right)\left(S_x + i S_y\right)^2 - 20 S_z^4\right) \\ \frac{3}{8}\sqrt{\frac{7}{2}} S_z \left(33 S_z^2 - 13\right)\left(S_x + i S_y\right)^4 \end{pmatrix} \tag{34}$$

$$-\nabla_{\mathbf{s}} Y_6^5(\mathbf{s}) = \begin{pmatrix} -\frac{3}{16}\sqrt{77} S_z \left(S_x + i S_y\right)^4 \left(6 i S_x S_y + S_x^2 - 5\left(S_y^2 + S_z^2\right)\right) \\ \frac{3}{16} i \sqrt{77} S_z \left(S_x + i S_y\right)^4 \left(6 i S_x S_y + 5 S_x^2 - S_y^2 + 5 S_z^2\right) \\ \frac{3}{16}\sqrt{77}\left(1 - 6 S_z^2\right)\left(S_x + i S_y\right)^5 \end{pmatrix} \tag{35}$$

$$-\nabla_{\mathbf{s}} Y_6^6(\mathbf{s}) = \begin{pmatrix} \frac{3}{16} i \sqrt{231}\left(S_x + i S_y\right)^5 \left(S_x S_y + i\left(S_y^2 + S_z^2\right)\right) \\ \frac{3}{16}\sqrt{231}\left(S_y - i S_x\right)^5 \left(i S_x S_y + S_x^2 + S_z^2\right) \\ \frac{3}{16}\sqrt{231} S_z \left(S_x + i S_y\right)^6 \end{pmatrix} \tag{36}$$

$$\tag{37}$$

We can tidy these up a bit to get

$$-\nabla_{\mathbf{s}} Y_2^0(\mathbf{s}) = \begin{pmatrix} 3S_x S_z^2 \\ 3S_y S_z^2 \\ 3S_z\left(S_z^2-1\right) \end{pmatrix} \tag{38}$$

$$-\nabla_{\mathbf{s}} Y_2^1(\mathbf{s}) = \begin{pmatrix} \sqrt{\frac{3}{2}}S_z\left(S_z^2-(S_x+iS_y)^2\right) \\ i\sqrt{\frac{3}{2}}S_z\left(S_z^2+(S_x+iS_y)^2\right) \\ \sqrt{\frac{3}{2}}\left(1-2S_z^2\right)(S_x+iS_y) \end{pmatrix} \tag{39}$$

$$-\nabla_{\mathbf{s}} Y_2^2(\mathbf{s}) = \begin{pmatrix} i\sqrt{\frac{3}{2}}(S_x+iS_y)\left(S_xS_y+i\left(S_y^2+S_z^2\right)\right) \\ \sqrt{\frac{3}{2}}(S_y-iS_x)\left(iS_xS_y+(S_x^2+S_z^2)\right) \\ \sqrt{\frac{3}{2}}S_z(S_x+iS_y)^2 \end{pmatrix} \tag{40}$$

$$-\nabla_{\mathbf{s}} Y_4^0(\mathbf{s}) = \begin{pmatrix} \frac{5}{2}S_xS_z^2\left(7S_z^2-3\right) \\ \frac{5}{2}S_yS_z^2\left(7S_z^2-3\right) \\ \frac{5}{2}S_z\left(S_z^2-1\right)\left(7S_z^2-3\right) \end{pmatrix} \tag{41}$$

$$-\nabla_{\mathbf{s}} Y_4^1(\mathbf{s}) = \begin{pmatrix} \frac{1}{4}\sqrt{5}S_z\left(-S_z^2(S_x+iS_y)(21S_x+iS_y)+3(S_x-iS_y)(S_x+iS_y)^3+4S_z^4\right) \\ \frac{1}{4}\sqrt{5}S_z\left(iS_z^2(S_x+iS_y)(S_x+21iS_y)+3(-S_y+iS_x)^3(S_x-iS_y)+4iS_z^4\right) \\ -\frac{1}{4}\sqrt{5}\left(28S_z^4-27S_z^2+3\right)(S_x+iS_y) \end{pmatrix} \tag{42}$$

$$-\nabla_{\mathbf{s}} Y_4^2(\mathbf{s}) = \begin{pmatrix} \frac{1}{2}\sqrt{\frac{5}{2}}(S_y-iS_x)\left(iS_z^2(S_x+iS_y)(8S_x+5iS_y)+S_y(S_x-iS_y)(S_x+iS_y)^2-6iS_z^4\right) \\ \frac{1}{2}\sqrt{\frac{5}{2}}(S_x+iS_y)\left(S_z^2(5S_x+8iS_y)(S_y-iS_x)+S_x(S_x+iS_y)^2(S_y+iS_x)-6iS_z^4\right) \\ \sqrt{\frac{5}{2}}S_z\left(7S_z^2-4\right)(S_x+iS_y)^2 \end{pmatrix} \tag{43}$$

$$-\nabla_{\mathbf{s}} Y_4^3(\mathbf{s}) = \begin{pmatrix} -\frac{1}{4}\sqrt{35}S_z(S_x+iS_y)^2\left(4iS_xS_y+S_x^2-3\left(S_y^2+S_z^2\right)\right) \\ \frac{1}{4}i\sqrt{35}S_z(S_x+iS_y)^2\left(4iS_xS_y-S_y^2+3(S_x^2+S_z^2)\right) \\ \frac{1}{4}\sqrt{35}\left(1-4S_z^2\right)(S_x+iS_y)^3 \end{pmatrix} \tag{44}$$

$$-\nabla_{\mathbf{s}} Y_4^4(\mathbf{s}) = \begin{pmatrix} \frac{1}{2}i\sqrt{\frac{35}{2}}(S_x+iS_y)^3\left(S_xS_y+i\left(S_y^2+S_z^2\right)\right) \\ \frac{1}{2}\sqrt{\frac{35}{2}}(-S_y+iS_x)^3\left(iS_xS_y+(S_x^2+S_z^2)\right) \\ \frac{1}{2}\sqrt{\frac{35}{2}}S_z(S_x+iS_y)^4 \end{pmatrix} \tag{45}$$

$$-\nabla_{\mathbf{s}} Y_6^0(\mathbf{s}) = \begin{pmatrix} \frac{21}{8} S_x S_z^2 \left(33S_z^4 - 30S_z^2 + 5\right) \\ \frac{21}{8} S_y S_z^2 \left(33S_z^4 - 30S_z^2 + 5\right) \\ \frac{21}{8} S_z \left(S_z^2 - 1\right) \left(33S_z^4 - 30S_z^2 + 5\right) \end{pmatrix} \tag{46}$$

$$-\nabla_{\mathbf{s}} Y_6^1(\mathbf{s}) = \begin{pmatrix} \frac{1}{8}\sqrt{\frac{21}{2}} S_z \, (-4S_z^4 (S_x + iS_y)(25S_x - 3iS_y) + 5S_z^2 (S_x - iS_y)(S_x + iS_y)^2 (17S_x + 3iS_y) - 5 (S_x - iS_y)^2 (S_x + iS_y)^4 \\ \frac{1}{8}\sqrt{\frac{21}{2}} S_z \, (4S_z^4 (3S_x - 25iS_y)(S_y - iS_x) + 5S_z^2 (S_x - iS_y)(S_x + iS_y)^2 (17S_y - 3iS_x) + 5i (S_x - iS_y)^2 (S_x + iS_y)^4 - \\ -\frac{1}{8}\sqrt{\frac{21}{2}} \left(198S_z^6 - 285S_z^4 + 100S_z^2 - 5\right)(S_x + iS_y) \end{pmatrix} \tag{47}$$

$$-\nabla_{\mathbf{s}} Y_6^2(\mathbf{s}) = \begin{pmatrix} \frac{1}{16}\sqrt{105}\,(S_x + iS_y) (64S_x S_z^4 (S_x + iS_y) - S_z^2 (S_x - iS_y)(S_x + iS_y)^2 (19S_x + 15iS_y) + iS_y (S_x - iS_y)^2 (S_x + iS_y)^3 \\ \frac{1}{16}\sqrt{105}\,(S_y - iS_x) (64iS_y S_z^4 (S_x + iS_y) - S_z^2 (S_x + iS_y)^2 \left(4iS_x S_y + 15S_x^2 + 19S_y^2\right) + S_x (S_x - iS_y)^2 (S_x + iS_y)^3 - \\ \frac{1}{16}\sqrt{105} S_z \left(99S_z^4 - 102S_z^2 + 19\right)(S_x + iS_y)^2 \end{pmatrix} \tag{48}$$

$$-\nabla_{\mathbf{s}} Y_6^3(\mathbf{s}) = \begin{pmatrix} \frac{3}{16}\sqrt{105} S_z (S_x + iS_y)^2 \left(-S_z^2 (S_x + iS_y)(13S_x + 5iS_y) + (S_x - iS_y)(S_x + iS_y)^2 (S_x + 3iS_y) + 8S_z^4\right) \\ -\frac{3}{16} i\sqrt{105} S_z (S_x + iS_y)^2 \left(-S_z^2 (S_x + iS_y)(5S_x + 13iS_y) + (S_x + iS_y)^2 \left(-2iS_x S_y + 3S_x^2 + S_y^2\right) - 8S_z^4\right) \\ -\frac{3}{16}\sqrt{105}\left(22S_z^4 - 15S_z^2 + 1\right)(S_x + iS_y)^3 \end{pmatrix} \tag{49}$$

$$-\nabla_{\mathbf{s}} Y_6^4(\mathbf{s}) = \begin{pmatrix} \frac{3}{8}\sqrt{\frac{7}{2}} (S_x + iS_y)^3 \left(S_z^2 (S_x + iS_y)(13S_x + 18iS_y) - 2iS_y (S_x - iS_y)(S_x + iS_y)^2 - 20S_z^4\right) \\ \frac{3}{8} i\sqrt{\frac{7}{2}} (S_x + iS_y)^3 \left(-S_z^2 (S_x + iS_y)(18S_x + 13iS_y) + 2S_x (S_x - iS_y)(S_x + iS_y)^2 - 20S_z^4\right) \\ \frac{3}{8}\sqrt{\frac{7}{2}} S_z \left(33S_z^2 - 13\right)(S_x + iS_y)^4 \end{pmatrix} \tag{50}$$

$$-\nabla_{\mathbf{s}} Y_6^5(\mathbf{s}) = \begin{pmatrix} -\frac{3}{16}\sqrt{77} S_z (S_x + iS_y)^4 \left(6iS_x S_y + S_x^2 - 5\left(S_y^2 + S_z^2\right)\right) \\ \frac{3}{16} i\sqrt{77} S_z (S_x + iS_y)^4 \left(6iS_x S_y - S_y^2 + 5(S_x^2 + S_z^2)\right) \\ \frac{3}{16}\sqrt{77}\left(1 - 6S_z^2\right)(S_x + iS_y)^5 \end{pmatrix} \tag{51}$$

$$-\nabla_{\mathbf{s}} Y_6^6(\mathbf{s}) = \begin{pmatrix} \frac{3}{16}\sqrt{231}\,(S_x + iS_y)^5 \left(iS_x S_y - \left(S_y^2 + S_z^2\right)\right) \\ \frac{3}{16}\sqrt{231}\,(S_y - iS_x)^5 \left(iS_x S_y + \left(S_x^2 + S_z^2\right)\right) \\ \frac{3}{16}\sqrt{231} S_z (S_x + iS_y)^6 \end{pmatrix} \tag{52}$$

(53)

REFERENCES

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